

# KMITL GPA Simulator - User Manual

A simple guide to help you use KMITL GPA Simulator.

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## Introduction

KMITL GPA Simulator helps you:

- **Calculate** your GPA from your courses
- **Plan** what grades you need to reach your goal
- **Test** how future grades will change your GPA
- **Export** your results as a PDF file

Website: <https://kmitl-gpa-simulator.pages.dev/>

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## Getting Started

1. Open the website in your browser
2. You will see four tabs:
  - **GPA Calculator** - Find your current GPA
  - **Goal Planner** - Set your GPA goal
  - **What-If Simulator** - Test different grades
  - **About** - Learn about grading
3. Click any tab to use it

**Note:** Your data is saved in your browser only. If you clear browser data or use another device, your courses will be lost.

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# GPA Calculator

Use this to calculate your current GPA.

## GPA Calculator

1. Enter your course name, credits, and grade, then click "Add" to add it to the list.
2. You can edit or delete any course after adding it.
3. Your GPA will be calculated automatically as you add courses.

Course Name

e.g., Introduction to Calculus

Credits

3

Grade

A

Add

| Course                         | Credits | Grade | Actions |        |
|--------------------------------|---------|-------|---------|--------|
| Introduction to Calculus       | 3       | A     | Edit    | Delete |
| Circuits and Electronics       | 4       | A     | Edit    | Delete |
| Elementary Systems Programming | 4       | B     | Edit    | Delete |
| Computer Programming           | 4       | B+    | Edit    | Delete |
| Linear Algebra                 | 3       | A     | Edit    | Delete |
| Probability and Statistics     | 3       | B+    | Edit    | Delete |

GPA: 3.64

Total Credits: 21

Export

Clear

Figure 1: GPA Calculator interface

## Add a Course

1. Go to the **GPA Calculator** tab
2. Enter your course info:
  - **Course Name:** Type the course name (e.g., "Calculus I")
  - **Credits:** Pick the credits (1-4)
  - **Grade:** Pick your grade (A, B+, B, C+, C, D+, D, F)
3. Click **Add Course**
4. Your GPA updates right away

## Edit a Course

1. Find the course in the table
2. Click the **Edit** button (pencil icon)
3. Change the info
4. Click **Update Course**

## Delete a Course

1. Find the course in the table
2. Click the **Delete** button (trash icon)
3. The course is removed and GPA updates

### **Clear All Courses**

1. Click **Clear All**
2. All courses are removed
3. GPA goes back to 0.00

### **Your Results**

- **Current GPA:** Your GPA score (0.00 - 4.00)
  - **Total Credits:** Sum of all credits
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# Goal Planner

Use this to check if your desired GPA is feasible.

**GPA Goal Planner**

1. Enter your current GPA and credits completed so far.  
2. Set your target GPA and total credits required for graduation.  
3. Click "Calculate" to see what grades you need to reach your target GPA.

**Current GPA:**  
3.29

**Credits Completed (affecting GPA):**  
137

**Total Credits Required for Graduation:**  
152

**Target GPA:**  
3.35

**Calculate Required Grades**

**GPA Goal Analysis**

|                  |                   |
|------------------|-------------------|
| Target GPA       | <b>3.35</b>       |
| Required Average | <b>3.90</b>       |
| Status           | <b>Achievable</b> |

To reach your target, you need to average **3.90** (mostly A grades) in your remaining **15** credits.

Figure 2: Goal Planner interface

## How to Use

1. Go to the **Goal Planner** tab
2. Enter your current info:
  - **Current GPA:** Your GPA now
  - **Credits Completed:** Credits you have finished
3. Enter your goal:
  - **Target GPA:** The GPA you want
  - **Total Credits:** Total credits needed to graduate
4. Click **Calculate**

## Your Results

- **Status:** Shows "Possible" or "Not Possible"
- **Required GPA:** The GPA you need in remaining courses
- **Grade Combinations:** Examples of grades that work

**Example**

| Input         | Value |
|---------------|-------|
| Current GPA   | 3.00  |
| Credits Done  | 60    |
| Target GPA    | 3.50  |
| Total Credits | 120   |

Result: You need **4.00** GPA in your next 60 credits.

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# What-If Simulator

Use this to see how future grades change your GPA.

### What-If GPA Simulator

1. Enter your current GPA and credits completed.  
2. Add future courses with expected grades to simulate.  
3. Click "Simulate" to see how future courses will affect your GPA.

**Calculation Type:**

Cumulative GPA (Show effect on overall GPA)

**Current GPA:**

3.5

**Credits Completed (affecting GPA):**

120

**Future Courses:**

| Course Name                    | Credits | Expected Grade |                      |
|--------------------------------|---------|----------------|----------------------|
| e.g., Introduction to Calculus | 3       | A              | <button>Add</button> |

| Course Name                | Credits | Grade | Actions                                       |
|----------------------------|---------|-------|-----------------------------------------------|
| Database Systems           | 3       | B+    | <button>Edit</button> <button>Delete</button> |
| Differential Equations     | 3       | A     | <button>Edit</button> <button>Delete</button> |
| Asian Study                | 3       | A     | <button>Edit</button> <button>Delete</button> |
| Probability and Statistics | 3       | A     | <button>Edit</button> <button>Delete</button> |
| Meditation for life        | 3       | A     | <button>Edit</button> <button>Delete</button> |

Simulate Future GPA

#### Simulation Results

|               |       |
|---------------|-------|
| Current GPA   | 3.50  |
| Projected GPA | 3.54  |
| Change        | +0.04 |

These courses will **maintain** your current GPA at around **3.54**.

Figure 3: What-If Simulator interface

## Two Modes

1. **Term GPA:** See your semester GPA and how it affects total GPA
2. **Cumulative GPA:** See your new total GPA directly

## How to Use

1. Go to the **What-If Simulator** tab
2. Enter your current info:

- **Current GPA:** Your GPA now
  - **Credits Completed:** Credits you have finished
3. Pick **Term** or **Cumulative** mode
  4. Add courses you plan to take:
    - Pick the **Grade** you expect
    - Pick the **Credits**
    - Click **Add Course**
  5. Click **Calculate**

## Your Results

### Term mode shows:

- **Term GPA:** GPA for new courses only
- **New GPA:** Your updated total GPA
- **GPA Change:** How much your GPA changed

### Cumulative mode shows:

- **Projected GPA:** Your new total GPA
- **GPA Change:** How much your GPA changed

## Tips

- Try different grades to see what happens
  - Focus more on courses that affect your GPA most
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## Exporting Your GPA Report

Save your GPA as a PDF file.

| Course Name                    | Credits | Grade |
|--------------------------------|---------|-------|
| Introduction to Calculus       | 3       | A     |
| Circuits and Electronics       | 4       | A     |
| Elementary Systems Programming | 4       | B     |
| Computer Programming           | 4       | B+    |
| Linear Algebra                 | 3       | A     |
| Probability and Statistics I   | 3       | B+    |

**GPA : 3.64    Total Credits : 21**

*Figure 4: PDF Export feature*

### How to Export

1. Go to **GPA Calculator** tab
  2. Add all your courses
  3. Click **Export as PDF**
  4. The PDF downloads to your device
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## Grading Scale

### Letter Grades

| Grade | Points | Meaning     |
|-------|--------|-------------|
| A     | 4.0    | Excellent   |
| B+    | 3.5    | Very Good   |
| B     | 3.0    | Good        |
| C+    | 2.5    | Fairly Good |
| C     | 2.0    | Fair        |
| D+    | 1.5    | Poor        |
| D     | 1.0    | Very Poor   |
| F     | 0.0    | Fail        |

### Special Grades

| Grade | Meaning        | Gets Credits? |
|-------|----------------|---------------|
| I     | Incomplete     | No            |
| S     | Satisfactory   | Yes           |
| U     | Unsatisfactory | No            |
| T     | Transfer       | Yes           |

### How GPA is Calculated

$$\text{GPA} = \text{Total Points} \div \text{Total Credits}$$

#### Example:

| Course  | Grade | Credits | Points                |
|---------|-------|---------|-----------------------|
| Math    | A     | 3       | $4.0 \times 3 = 12.0$ |
| Physics | B+    | 3       | $3.5 \times 3 = 10.5$ |
| English | B     | 2       | $3.0 \times 2 = 6.0$  |

**Total Points:**  $12.0 + 10.5 + 6.0 = 28.5$  **Total Credits:**  $3 + 3 + 2 = 8$  **GPA:**  $28.5 \div 8 = 3.56$

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# Mobile Usage

This app works on phones and tablets.

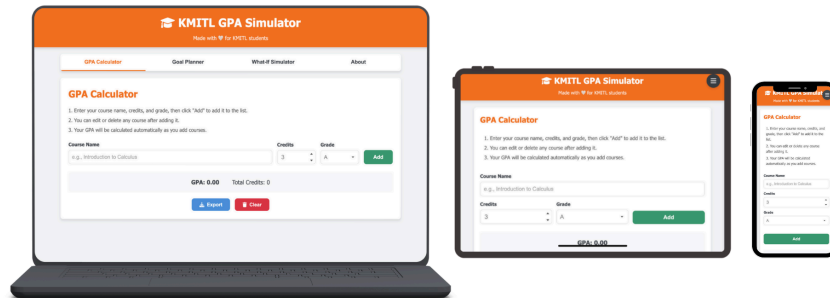


Figure 5: Responsive design across devices (desktop, tablet, mobile)

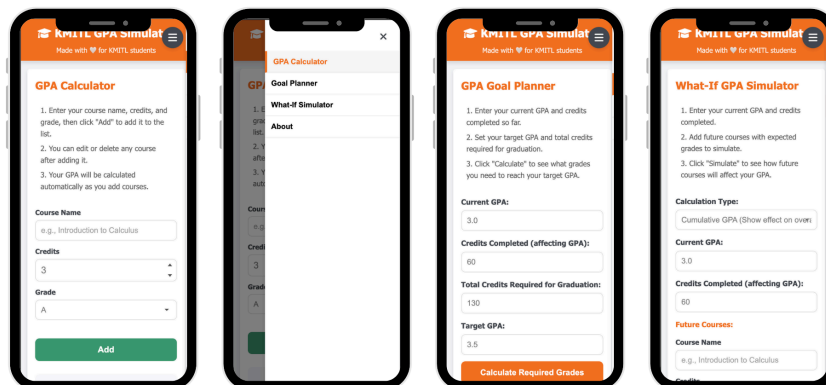


Figure 6: Mobile interface with hamburger menu

## Using on Mobile

1. Tap the **menu icon** ( $\equiv$ ) at the top right
2. Pick the tab you want
3. Tap outside or press X to close the menu

## Need Help?

If you have problems or for further contribution:

- Contact the developer at [phyoethikhine143@gmail.com](mailto:phyoethikhine143@gmail.com)
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## Reference

- [KMITL Academic Regulations \(ข้อบังคับสถาบันว่าด้วยการศึกษาระดับปริญญาตรี พ.ศ. 2565\)](#)
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*Developed by Phyo Thi Khaing (65011466) | SIIE Scholarship Student Contributed to KMITL | 2026*